BANOTH ANUSHA

Indian Institute of Technology, Goa

B.Tech (Graduated 2026), **Computer Science and Engineering**
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Education

B.Tech in Computer Science and Engineering, Indian Institute of Technology Goa	CGPA: **5.03**/10	2022 – 2026 **(Graduated)**
Class 12, State Board, T.T.W. Residential Junior College of Excellence (Girls), Warangal	Aggregate: **89.4**%	2020 – 2022
Class 10, State Board, T.S. TWRES School for Girls, Kuravi, Mahabubabad District	CGPA: **9.8**/10	2019 – 2020

Professional Summary

- Computer Science graduate from IIT Goa who likes taking an AI idea all the way to a working product. Built a fake news classifier (98% test accuracy) with a live Streamlit demo, a **YOLOv8 + ByteTrack** player tracking pipeline over 3,500+ frames, and a full-stack **RAG assistant** that answers questions from PDFs with page-level citations. Comfortable with **Python, C++, and SQL**. Looking for an entry-level **AI/ML Engineer or Software Engineer** role.

Research Experience

Bachelor's Thesis Project (BTP) – Indian Institute of Technology Goa (2025 – 2026)

- Carried out two independent research projects under faculty supervision: **fake news classification** (NLP) and **multi-player detection and tracking** in cricket footage (computer vision).
- Owned both projects from start to finish: data preparation, model development, evaluation, analytics, and visualization, and deployed the NLP model as a public web demo.

Projects

Fake News and Misinformation Detector [GitHub] [Live Demo] (Nov 2025 – Feb 2026)

- Built an end-to-end **NLP classification pipeline** for REAL/FAKE news detection using **TF-IDF** and **Logistic Regression**, achieving **98% accuracy** on the test dataset.
- Preprocessed text and extracted features with **Pandas, NumPy, and TF-IDF**; evaluated reliability using a **Confusion Matrix, ROC Curve, and Precision–Recall Curve**.
- Deployed the model as an interactive **Streamlit web application** for real-time news classification.
- **Tech Stack:** Python, Scikit-learn, TF-IDF, Logistic Regression, Streamlit, Pandas, NumPy, Matplotlib

Sports Player Detection and Tracking Pipeline [GitHub] (Nov 2025 – Apr 2026)

- Built an end-to-end **YOLOv8 + ByteTrack** pipeline for multi-person detection and persistent player tracking across **3,500+ frames** of ICC T20 cricket footage.
- Maintained **65+ unique player IDs** through occlusion and camera motion, and added **speed estimation (km/h)**, motion trails, heatmaps, and a live HUD.
- Automated analysis with **four performance graphs, CSV exports, and 117 automated screenshots**; benchmarked ByteTrack against BoT-SORT.
- **Tech Stack:** Python, YOLOv8 (Ultralytics), ByteTrack, OpenCV, NumPy, Pandas, Matplotlib, Supervision

Enterprise RAG Knowledge Assistant [GitHub] (2026)

- Built a full-stack document Q&A app where users upload PDFs and get **streaming answers with page-level citations**, using **FastAPI, Next.js (TypeScript), and Docker Compose**.
- Implemented **hybrid retrieval** that merges **pgvector** similarity search with PostgreSQL full-text search using **Reciprocal Rank Fusion**, then reranks with a **cross-encoder**; added a relevance cutoff and server-side **citation validation**.
- Designed **asynchronous ingestion** with **Celery and Redis**; secured the service with **JWT auth**, per-user document isolation, and rate limits. Works with OpenAI-compatible LLMs (tested with Groq) or local **Ollama**.
- **Tech Stack:** Python, FastAPI, PostgreSQL/pgvector, Celery, Redis, Next.js, TypeScript, Docker

Skills

Programming:	Python, C++, SQL, TypeScript
AI and Machine Learning:	Machine Learning, Deep Learning, NLP, Computer Vision, RAG, Object Detection and Tracking, Scikit-learn, YOLOv8, OpenCV, TF-IDF
Backend and Web:	FastAPI, Next.js, PostgreSQL, pgvector, Redis, Celery, Docker, REST APIs, JWT Authentication, Streamlit
Core CS:	Data Structures and Algorithms, Operating Systems, Computer Networks
Relevant Coursework:	Machine Learning, Artificial Intelligence, Probability and Statistics, Linear Algebra, Algorithm Design, Algebraic Algorithms, Discrete Structures
Tools:	Git, GitHub, Linux, VS Code

Positions of Responsibility

Event Coordinator, Cultrang Fest:	Planned and coordinated fest activities, managed team tasks and on-ground logistics at IIT Goa.
Team Member, Motorsports Club:	Supported vehicle system design through analysis and problem solving in a team at IIT Goa.

Certifications

- Completed *The Complete Full-Stack Web Development Bootcamp* (61.5 hrs) on Udemy, taught by Dr. Angela Yu. July 2025

Extracurricular Activities

Sports and Games	Badminton, Volleyball, Chess
Events	Participated in cultural fests and national day celebrations at IIT Goa